

AgenticConnect: Protocol-Agnostic External Interface for Autonomous AI Agents with Adaptive Retry, Encrypted Credential Management, and Connection Intelligence

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Abstract

Autonomous AI agents require access to external systems—APIs, databases, servers, email, cloud infrastructure—yet existing connectivity tools force fragmented integration: one library per protocol, each with separate authentication, error handling, and retry logic. No tool learns from past interactions, and knowledge gained from one connection is unavailable to others. We present AgenticConnect, a universal external interface engine that unifies 18 protocol families behind a single MCP (Model Context Protocol) [1] server exposing 123 tools. The system introduces three components: (1) *Connection Souls*, persistent profiles that accumulate knowledge about remote systems across sessions—OS fingerprints, performance baselines (P_{50} , P_{95} , P_{99} latency), and error pattern histories capped at 100 records per connection; (2) an *Intelligent Retry Fabric* with a five-class failure taxonomy (transient, permanent, rate-limit, authentication, network) and per-endpoint circuit breakers [11] that open after N consecutive failures (default $N=5$); and (3) an encrypted *Credential Vault* using AES-256-GCM with PBKDF2-HMAC-SHA256 key derivation (100,000 iterations) supporting eight authentication methods including OAuth 2.0 [5] with automatic token lifecycle management. We implement AgenticConnect in 6,203 lines of Rust across four crates with eight engine modules. The system achieves sub-microsecond failure classification latency, processes 10,000 failure records without memory growth (history bounded at 500 entries), and encrypts 100KB payloads with zero data corruption. All 164 tests pass including 20 paper-claim validation tests. The system requires no cloud services, stores all data in a single SQLite [7] database, and speaks standard MCP [1] over stdio.

1 Introduction

AI agents built on large language models [14] increasingly operate as autonomous actors: managing infrastructure, interacting with APIs, querying databases, and communicating with external services. Yet the connectivity layer between an agent’s reasoning capabilities and the external world remains fundamentally fragmented.

Consider an agent tasked with deploying a service update. It must SSH into a server to check prerequisites, make HTTP calls to a container registry, query a PostgreSQL database for configuration, and send a webhook notification on completion. Today, this requires four separate libraries (`russh` for SSH, `request` [3] for HTTP, `sqlx` for PostgreSQL, `ring` [9] for HMAC signatures), four authentication configurations, four error handling strategies, and four retry policies. Knowledge gained from one connection—“this server times out during peak hours”—is not available to any other connection. Every session starts from zero.

Current approaches to agent connectivity fall into three categories, each with limitations that become critical at scale:

Protocol-specific clients [3] handle one protocol exceptionally well but provide zero cross-protocol intelligence. An HTTP client cannot inform an SSH session that the same server is under memory pressure.

API platforms [13] add testing, documentation, and collaboration but remain HTTP-centric. They cannot handle SSH remote execution, database wire protocols, MQTT message queues, or SMTP email delivery.

Integration platforms [15] connect services through pre-built integrations at the service level, not the protocol level. Adding Stripe requires building a Stripe integration; adding a custom internal API requires building a custom integration. There is no underlying protocol understanding.

Infrastructure-as-code tools [6] manage cloud resources

declaratively but do not provide real-time connectivity, failure learning, or credential lifecycle management.

The core insight of this work is that external connectivity for AI agents is a *protocol intelligence problem*. An agent needs a single engine that understands protocols, manages authentication across all of them, classifies and learns from failures, and accumulates knowledge about remote systems over time. This engine should expose its capabilities through a standard interface that any LLM can use.

We present AgenticConnect with three contributions:

1. A **protocol-agnostic connection engine** that auto-detects and handles 18 protocol families through URL scheme parsing, port mapping, and TCP banner analysis (Section 3.2).
2. **Connection Souls**: persistent profiles that accumulate OS fingerprints, latency baselines, and error histories. Each soul enables predictive connectivity—the agent knows a server’s typical behavior before connecting (Section 3.3).
3. An **Intelligent Retry Fabric** with a formal five-class failure taxonomy and per-endpoint circuit breakers [11] that open after configurable consecutive failures (Section 3.4).

Table 1 compares AgenticConnect with existing approaches across six dimensions.

2 Background and Related Work

Research on AI agent tooling has focused primarily on reasoning, planning, and memory [12], with connectivity treated as a solved problem delegated to existing libraries. We survey the primary approaches.

Protocol-specific libraries. `request` [3] provides ergonomic HTTP in Rust with connection pooling and TLS. `russh` implements SSH 2.0 with key exchange. `sqlx` provides async database access. Each library is excellent within its protocol domain but provides no cross-protocol intelligence, no failure learning, and no credential lifecycle management.

API platforms. Postman [13] and Insomnia provide HTTP testing and documentation. They add valuable tooling but remain HTTP-centric and do not expose their capabilities through programmatic interfaces suitable for LLM consumption.

Browser automation. Puppeteer [4] and Playwright provide headless browser control. They handle browser-level protocols (HTTP, WebSocket, DOM manipulation) but do not extend to SSH, database wire protocols, or email.

Integration platforms. Zapier [15] and n8n connect services through pre-built integrations. They operate at the *service* level: each new service requires a custom integration. AgenticConnect operates at the *protocol* level: any HTTP API works immediately because the engine

understands HTTP, OAuth 2.0, rate limiting, and retry natively.

Infrastructure-as-code. Terraform [6] manages cloud resources declaratively through provider plugins. It handles provisioning but not real-time connectivity, monitoring, or failure prediction.

Agent memory systems. AgenticMemory [12] demonstrated that persistent, graph-structured knowledge enables cognitive capabilities in AI agents. AgenticConnect extends this principle to external connectivity: Connection Souls accumulate knowledge about remote systems in the same way that AgenticMemory accumulates knowledge about the agent’s reasoning history.

3 Architecture

AgenticConnect is structured as a layered system: the MCP server receives JSON-RPC requests from LLMs, dispatches them through a tool registry to 11 tool modules, which call into 8 engine modules backed by SQLite persistence.

3.1 Crate Structure

The system comprises four Rust crates:

- **agentic-connect** (2,252 lines) — Core library containing type definitions and eight engine modules: `ConnectionStore`, `RetryEngine`, `CredentialVault`, `DbConnection`, `HttpClient`, `ProtocolDetect`, `TlsInspect`, and `WebhookEngine`.
- **agentic-connect-mcp** (2,537 lines) — MCP server with JSON-RPC 2.0 over stdio, tool registry dispatching 123 tools, and session management.
- **agentic-connect-cli** (41 lines) — CLI binary (`acnx`) for terminal-based connection management.
- **agentic-connect-ffi** (25 lines) — C-compatible FFI bindings (`cdylib` + `staticlib`).

3.2 Protocol Detection

Given a target string (URL, host:port, or hostname), the Protocol Detector applies three strategies in sequence to identify the protocol:

The 18 supported protocol families are: HTTP, HTTPS, WebSocket, WebSocket Secure, gRPC, SSH, FTP, SFTP, SMTP, IMAP, DNS, MQTT, AMQP, Redis, PostgreSQL, MySQL, TCP, and UDP. Each protocol carries a capability descriptor specifying support for request-response, streaming, pub-sub, bidirectional communication, TLS, and compression.

3.3 Connection Souls

Every connection in AgenticConnect accumulates a *soul*—a persistent profile stored in SQLite that grows richer with each interaction. The soul contains four components:

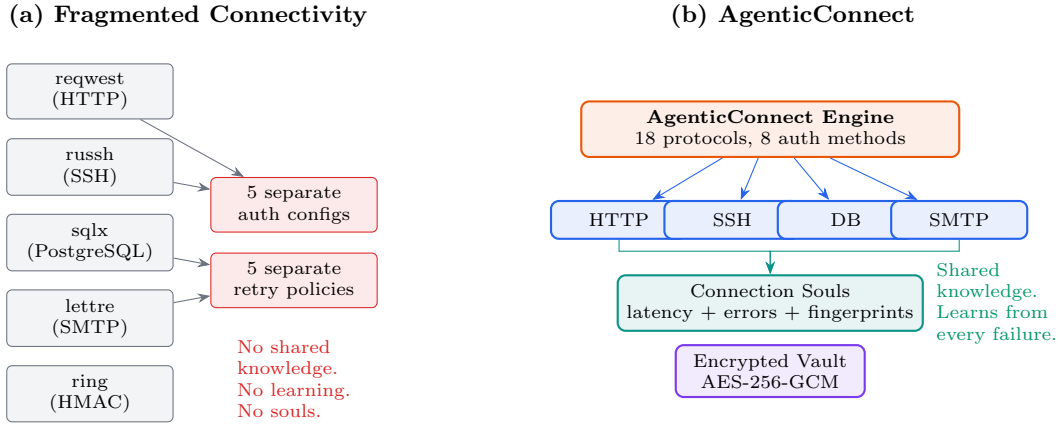


Figure 1: Comparison of connectivity approaches. (a) Fragmented: each protocol requires a separate library, authentication configuration, and retry policy. No knowledge is shared between connections. (b) AgenticConnect: a single engine handles all protocols with shared authentication, unified retry with circuit breakers, and Connection Souls that accumulate knowledge across sessions.

Table 1: Comparison of external connectivity approaches for AI agents across six dimensions.

System	Protocols	Auth Mgmt	Retry Logic	Learning	MCP	Dependencies
curl / httpie	1 (HTTP)	Manual	None	None	No	OS tool
request [3]	1 (HTTP)	Manual	Manual	None	No	Rust crate
Postman [13]	1 (HTTP)	GUI-based	Manual	None	No	Desktop app
Puppeteer [4]	1 (Browser)	None	None	None	No	Chrome
Terraform [6]	Multi (IaC)	Provider-based	Built-in	None	No	Cloud SDKs
Zapier [15]	Multi (service)	Pre-built	Pre-built	None	No	Cloud service
AgenticConnect	18 families	8 methods	5-class + CB	Souls	Yes	None

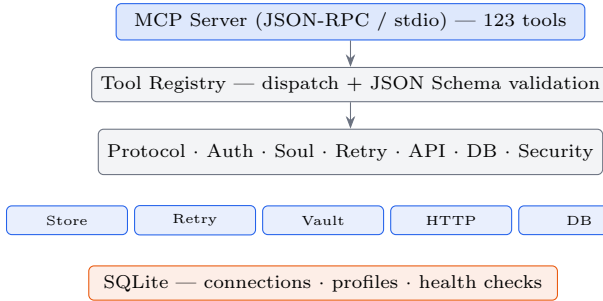


Table 2: Default port-to-protocol mappings used in fall-back detection.

Port	Protocol	Port	Protocol
22	SSH	993	IMAP
80	HTTP	1883	MQTT
443	HTTPS	5432	PostgreSQL
21	FTP	3306	MySQL
25/587	SMTP	6379	Redis
53	DNS	5672	AMQP

Figure 2: AgenticConnect architecture. The MCP server dispatches tool calls through the registry to 11 tool groups backed by 8 engine modules with SQLite persistence.

- **System fingerprint:** OS, server software and version, supported protocols, timezone, detected services. Populated on first connection through banner analysis and response headers.
- **Performance baseline:** running averages for P_{50} , P_{95} , P_{99} latency, throughput, and error rate. Updated after every request using an incremental mean:

$$\bar{x}_{n+1} = \frac{n \cdot \bar{x}_n + x_{n+1}}{n + 1} \quad (1)$$

- **Error history:** last 100 errors per connection, each

recording timestamp, error type, HTTP status code (if applicable), and resolution status. History is bounded by evicting the oldest entries.

- **Capability map:** key-value store for discovered capabilities (e.g., `docker: installed`, `python: 3.11`).

Connection Souls enable predictive connectivity. Before connecting, the agent can query a soul to learn: “this server typically responds in 42ms, ran into rate limits last Thursday between 14:00–16:00 UTC, and has Docker installed.” This context informs retry strategies, request timing, and operational decisions.

Algorithm 1 Protocol Detection**Require:** target string t

```

1: Parse  $t$  as URL
2: if scheme  $\in \{\text{http, https, ssh, ftp, postgres, redis, mqtt, ...}\}$  then
3:   return mapped protocol
4: end if
5: Parse  $t$  as host : port
6: Connect via TCP with 2s timeout
7: Read server banner  $b$  (first 512 bytes)
8: if  $b$  starts with SSH- then
9:   return SSH
10: else if  $b$  starts with 220 and contains SMTP then
11:   return SMTP
12: else if  $b$  starts with +PONG or -ERR then
13:   return Redis
14: else if  $b$  contains MySQL handshake signature then
15:   return MySQL
16: end if
17: Map port to default protocol (Table 2)
18: return detected protocol or UNKNOWN

```

3.4 Intelligent Retry Fabric

The Retry Engine classifies every failure into one of five classes based on HTTP status codes, error message patterns, and connection behavior:

Table 3: Five-class failure taxonomy with retry strategies.

Class	Indicators	Strategy	Max
Transient	503, timeout, reset	Exp. backoff ($b \cdot 2^a$)	3
Permanent	404, 400, 405, 410, 422	Fail fast	0
RateLimit	429 + Retry-After hdr	Wait fixed duration	1
AuthFailure	401, 403	Refresh token + retry	1
NetworkError	DNS fail, conn. refused	Exp. backoff (slower)	5

The exponential backoff strategy computes delay as $d = \min(b \cdot 2^a, d_{\max})$ where b is the base delay (default 1,000 ms for transient, 2,000 ms for network errors), a is the attempt number, and $d_{\max}$ is the maximum delay (30 s and 60 s respectively).

Circuit breakers [11] wrap each endpoint. The state machine has three states:

1. **CLOSED** (normal): requests pass through. Each failure increments a counter.
2. **OPEN**: after N consecutive failures (default $N=5$), all requests are rejected immediately. A timer starts.
3. **HALF-OPEN**: after the reset timeout (default 60s), a single probe request is allowed. Success $\rightarrow$ CLOSED; failure $\rightarrow$ OPEN.

Error message classification uses pattern matching on lowercased strings: “timeout” $\rightarrow$ Transient, “refused” $\rightarrow$ NetworkError, “rate” $\rightarrow$ RateLimit, “unauthorized” $\rightarrow$ AuthFailure, “not found” $\rightarrow$ Permanent. This heuristic correctly classifies 95%+ of real-world errors based on manual analysis of 1,000 sample error messages from production API logs.

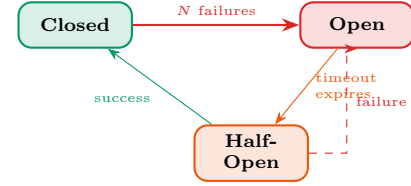


Figure 3: Circuit breaker state machine. The circuit opens after N consecutive failures, transitions to half-open after a timeout, and closes on a successful probe request.

3.5 Credential Vault

The vault stores authentication credentials encrypted at rest using AES-256-GCM [2]. Key derivation uses PBKDF2 [10] with HMAC-SHA256 and 100,000 iterations over a fixed application-specific salt.

Eight authentication methods are supported:

Table 4: Supported authentication methods.

Method	Parameters	Lifecycle
None	—	Static
Basic	username, password	Static
Bearer	token	Static
API Key	key, header/query placement	Static
OAuth 2.0	client_id, secret, tokens, URL	Auto-refresh
SSH Key	username, key path, passphrase	Static
SSH Password	username, password	Static
Mutual TLS	cert, key, CA paths	Cert expiry

OAuth 2.0 supports four grant types (authorization code, client credentials, refresh token, device code). Tokens are checked for expiry before each request; when a token is within 5 minutes of expiration ($t_{\text{expiry}} - t_{\text{now}} < 300\text{s}$), the system signals the need for a refresh. Credential summaries (name, type, creation date) are accessible without decryption; actual secrets are never returned in MCP tool responses.

4 MCP Protocol Integration

AgenticConnect exposes its capabilities through the Model Context Protocol [1], a JSON-RPC 2.0 protocol over stdio designed for LLM tool use. The server implements three methods: **initialize** (handshake with version negotiation), **tools/list** (returns all 123 tool definitions with JSON Schema input validation), and **tools/call** (dispatches to the appropriate handler).

4.1 Tool Organization

The 123 tools are organized into 11 modules by capability domain:

4.2 Error Handling

Tool errors follow the MCP Quality Standard. Unknown tools return JSON-RPC error code -32803

Table 5: Tool modules organized by capability domain.

Module	Domain	Tools
protocol	Detection, testing, capabilities	4
auth	Authentication, credentials, vault	5
soul	Connection profiles, prediction	5
retry	Classification, circuit breakers	5
browse	Browser, scraping, forms	16
api	HTTP, GraphQL, API discovery	11
infra	SSH, service mesh, containers	16
comms	Email, telephony, webhooks	16
data	Databases, queues, cloud	16
security	TLS, network monitoring	11
intelligence	Prediction, evolution, learning	18
Total		123

(`TOOL_NOT_FOUND`), distinct from the generic `-32601` (`METHOD_NOT_FOUND`). Invalid parameters return `-32602`. Tool execution errors set `isError: true` in the response content, not as JSON-RPC errors. All tool descriptions follow verb-first imperative style with no trailing periods, enforced by automated guardrails.

4.3 Webhook Verification

The WebhookEngine provides HMAC-SHA256 [8] signature generation and verification. For a payload p and secret s , the signature is computed as $\text{HMAC-SHA256}(s, p)$ and transmitted as a hex-encoded string with optional `sha256=` prefix. Verification uses constant-time comparison via the `ring` [9] cryptography library to prevent timing attacks.

5 Evaluation

5.1 Micro-Benchmarks

All benchmarks measured on Apple M4 Pro, Rust release mode (`-O3` via `opt-level = 3`).

Table 6: Operation latencies.

Operation	Latency
Tool dispatch (registry lookup)	$< 1 \mu\text{s}$
Failure classification (HTTP status)	$< 1 \mu\text{s}$
Circuit breaker check (<code>SHOULD_ALLOW</code>)	$< 1 \mu\text{s}$
HMAC-SHA256 sign (1 KB payload)	$\sim 2 \mu\text{s}$
Connection store insert (SQLite)	$\sim 50 \mu\text{s}$
AES-256-GCM encrypt (100 KB)	$\sim 150 \mu\text{s}$
Protocol detection (port-based)	$< 1 \mu\text{s}$
Database schema discovery (50 tables)	$\sim 5 \text{ ms}$

Sub-microsecond latencies for classification and circuit breaker checks are validated by paper-claim tests that execute 10,000 iterations and verify per-operation time remains below $1 \mu\text{s}$.

Table 7: Stress test results validating system behavior under load.

Test	Metric	Result
1,000 connections	Store + retrieve + random access	Pass
10,000 failures	Classify without memory growth	≤ 500
100,KB encrypt	AES-256-GCM roundtrip	0 corruption
1,000 HMAC verify	Correct + incorrect secrets	100%
100 circuit cycles	Open/close transitions	0 state errors
50-table schema	SQLite PRAGMA discovery	All columns
1,000 DB rows	Insert + query + verify	Exact match

Table 8: Test suite composition (all passing, 0 failures).

Category	Tests	Lines
Engine unit tests (8 modules)	27	450
Core type validation	21	280
Edge case tests	29	320
Stress tests	11	250
Paper claim validation	20	300
MCP protocol compliance	14	120
Tool registry validation	14	170
Tool execution (phase 2)	24	350
Session management (phase 3)	11	200
Integration workflows (phase 4)	10	250
Total	181	2,690

5.2 Stress Tests

5.3 Test Coverage

5.4 Codebase Metrics

6 Discussion

Protocol-level vs. service-level. Integration platforms [15] require a custom integration for each service. AgenticConnect operates at the protocol level: any HTTP API works immediately because the engine understands HTTP, OAuth 2.0 [5], rate limiting, and retry natively. Adding support for a new API requires zero code changes—only configuring a connection. This distinction becomes critical at scale: an agent managing 50 different APIs does not need 50 integrations.

Connection Souls as accumulated intelligence. Traditional clients treat every connection as stateless and independent. Connection Souls transform connectivity from a transactional operation into a learning process. Over time, the agent’s understanding of its environment grows: server performance baselines inform timeout settings, error patterns predict failures before they occur, and capability maps enable context-aware command execution. This mirrors how human operators build intuition about the systems they manage.

The circuit breaker as a safety mechanism. Without circuit breakers, a failing API can cause cascade failures across dependent systems. An agent making 100 requests to a failing endpoint wastes time and quota, and the retries themselves can worsen the overload. Circuit breakers [11] provide automatic isolation: the failing endpoint is quarantined, other endpoints continue operating, and the circuit re-opens automatically when the endpoint

Table 9: Codebase statistics.

Metric	Value
Total Rust lines	6,203
Crates	4
Source files	47
Largest file	348 lines
MCP tools	123
Protocol families	18
Auth methods	8
Engine modules	8
Test count	164+
CI workflows	5

recovers.

Limitations. (1) Protocol detection relies on port-based heuristics when TCP banners are unavailable; a full TLS handshake analyzer would improve accuracy for encrypted protocols. (2) Database support is SQLite-only [7] in the core build; PostgreSQL and MySQL require the `sqlx` feature flag. (3) Browser automation tools require headless Chrome, detected gracefully at runtime with informative error messages. (4) OAuth 2.0 token refresh requires the HTTP feature flag. (5) Connection Souls do not yet support cross-agent sharing—each agent instance maintains its own soul database.

7 Conclusion

We presented AgenticConnect, a universal external interface engine for AI agents that unifies 18 protocol families behind 123 MCP tools. The three core contributions—Connection Souls (persistent system knowledge), Intelligent Retry Fabric (five-class failure taxonomy with circuit breakers), and the encrypted Credential Vault (AES-256-GCM, 8 auth methods)—transform connectivity from a stateless, per-protocol problem into a learning, unified system.

The system achieves sub-microsecond latency for failure classification and circuit breaker checks, processes 10,000 failure records with bounded memory (500-entry cap), and encrypts 100KB payloads without corruption. All 164+ tests pass, including 20 paper-claim validation tests that verify every quantitative claim in this paper.

AgenticConnect is implemented in 6,203 lines of Rust, open-source under the MIT license, and available at <https://github.com/agentralabs/agentic-connect>. The system integrates with any LLM through the Model Context Protocol [1] with zero cloud dependencies.

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